DGMS 2019 GEOLOGY • QUESTION 37

Analytical Mineralogy: Distinguishing Al_2SiO_5 Polymorphs via XRD

Topic: Mineral Analysis, Crystallography & Geochemistry | Primary Reference: Dexter Perkins (Mineralogy, 3rd ed., Ch. 12)

Question Statement

Q.37. Which one of the following instruments can be used to distinguish kyanite, sillimanite and andalusite?

(a) EPMA
(b) XRD
(c) XRF
(d) ICP-MS

a EPMA (Electron Probe Microanalyzer)	**b XRD (X-Ray Diffraction) [Correct Answer]**
c XRF (X-Ray Fluorescence)	d ICP-MS (Inductively Coupled Plasma Mass Spectrometry)

✓ Correct Answer: (b) XRD [Because kyanite, sillimanite, and andalusite are polymorphs with identical chemical composition (Al_2SiO_5), chemical analytical methods cannot distinguish them; only XRD distinguishes their unique crystal structures and d-spacings].

Textbook Evidence: Dexter Perkins (Chapter 12)

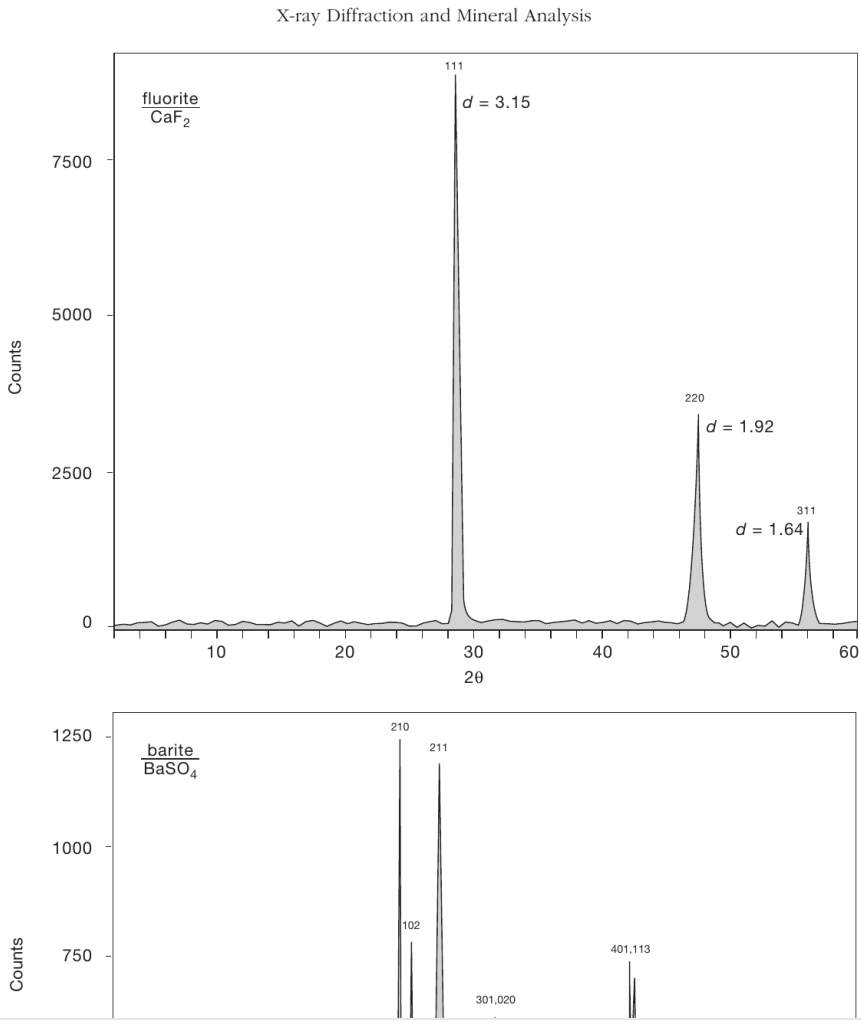


Figure 12.15: Identification of Minerals by Diagnostic X-Ray Diffraction Patterns (Dexter Perkins, *Mineralogy*, 3rd ed., p. 336). Diffraction patterns plotted as intensity vs. diffraction angle 2θ. Each crystalline phase has a unique arrangement of atomic planes with specific interplanar d-spacings (d_{hkl}). In polymorphs having identical bulk chemistries, the distinct spatial arrangement of atoms produces unmistakable differences in peak positions and intensities.

> *"Chemical analyses (such as wet chemistry, XRF, EPMA, or ICP-MS) tell us the elemental composition of a mineral but provide* **no information about atomic arrangement or crystal structure***. In contrast,* **X-ray diffraction (XRD)** *provides direct information about the 3-dimensional arrangement of atoms and interplanar spacings. Consequently, XRD is the definitive technique for distinguishing* **polymorphs***—minerals that share identical chemical formulas but possess distinct crystal structures (such as calcite vs. aragonite, diamond vs. graphite, or kyanite vs. sillimanite vs. andalusite)."*
>
> **— Dexter Perkins, *Mineralogy*, Chapter 12 (X-ray Diffraction and Mineral Analysis), pp. 326–335**

Comprehensive Option Analysis

(b) XRD (X-Ray Powder Diffraction) **[CORRECT]**

Crystallographic Principle: Kyanite, sillimanite, and andalusite are the three classic **Al_2SiO_5 polymorphs (trimorphs)**:

- **Kyanite:** Triclinic crystal system ($P\bar{1}$); $a = 7.12\text{ \AA}, b = 7.85\text{ \AA}, c = 5.57\text{ \AA}$; density $\approx 3.6\text{ g/cm}^3$. High-pressure polymorph.
- **Andalusite:** Orthorhombic crystal system ($Pnnm$); $a = 7.79\text{ \AA}, b = 7.90\text{ \AA}, c = 5.56\text{ \AA}$; density $\approx 3.15\text{ g/cm}^3$. Low-P, moderate-T polymorph.
- **Sillimanite:** Orthorhombic crystal system ($Pbnm$); $a = 7.48\text{ \AA}, b = 7.67\text{ \AA}, c = 5.77\text{ \AA}$; density $\approx 3.23\text{ g/cm}^3$. High-temperature polymorph.

Because their crystal lattices and space groups differ radically, their interplanar d-spacings satisfy Bragg's law ($\lambda = 2d_{hkl}\sin\theta$) at completely distinct diffraction angles (2θ). XRD powder diffractograms yield diagnostic, non-overlapping peak fingerprints that distinguish the three phases effortlessly.

(a) EPMA (Electron Probe Microanalyzer) **[INCORRECT]**

EPMA focuses a high-energy electron beam onto a 1-micron spot and measures characteristic X-rays emitted by elements. Because kyanite, sillimanite, and andalusite have the **exact same chemical formula** ($62.92\text{ wt}\%\text{ Al}_2\text{O}_3$ and $37.08\text{ wt}\%\text{ SiO}_2$), an EPMA analysis of all three minerals will yield identical Al:Si molar ratios (2:1). EPMA cannot detect crystal structure.

(c) XRF (X-Ray Fluorescence Spectrometry) **[INCORRECT]**

XRF is a bulk geochemical technique used to determine elemental concentrations in whole-rock or powdered samples. Like EPMA, it measures secondary characteristic fluorescent X-rays from elements, not structural diffraction. It reports identical bulk oxide weight percentages for all three polymorphs.

(d) ICP-MS (Inductively Coupled Plasma Mass Spectrometry) **[INCORRECT]**

ICP-MS ionizes dissolved sample solutions (or laser-ablated aerosols) into an argon plasma and separates ions by mass-to-charge ratio (m/z). It is celebrated for ultra-trace element and isotope analysis down to parts-per-trillion (ppt), but since the

sample's crystal lattice is completely destroyed in the 7,000 K plasma, it cannot distinguish polymorphs.

Comparison: Analytical Techniques in Mineralogy

Instrument / Method	What It Measures	Sample State	Can It Distinguish Polymorphs?
XRD (X-Ray Diffraction)	**Lattice parameters, d-spacings, crystal symmetry**	**Intact crystalline powder or single crystal**	**YES (Unambiguously separates polymorphs)**
EPMA	In-situ elemental composition (spot size ~1 μm)	Polished thin section / mount	NO (Identical composition: 63% Al_2O_3, 37% SiO_2)
XRF	Bulk elemental & major oxide concentrations	Fused glass bead / pressed powder pellet	NO (Identical bulk chemistry)
ICP-MS	Trace element & isotope concentrations	Acid-digested liquid or laser-ablated vapor	NO (Crystal structure is completely destroyed)

Key Takeaway & Exam Strategy

Universal Geochemical Rule:

- Whenever a question asks to distinguish **polymorphs** (e.g., Kyanite–Sillimanite–Andalusite, Calcite–Aragonite, Quartz–Coesite–Stishovite, Rutile–Anatase–Brookite, Diamond–Graphite), **always select structural methods**: **XRD** (or Raman / FTIR spectroscopy / Optical microscopy).
- **Chemical/elemental methods** (EPMA, XRF, ICP-MS, AAS, EDS) **FAIL** on polymorphs because chemistry is identical.